

Charges and Charging, and Coulomb's Law

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1 Charging Introduction

These are my notes on charges and charging for Physics!

Charges are fundamental properties for particles, just like **masses** and **spins**.

2 Fundamental Properties of Particles

There are three particles that matter for charges and charging.

Particle	Charge (in e)	Charge (in Coulombs (C))
Proton	$1e$	$1.6 \times 10^{-19} \text{ C}$
Electron	$-1e$	$-1.6 \times 10^{-19} \text{ C}$
Neutron	$1e$	$1.6 \times 10^{-19} \text{ C}$

Charges are usually sent in multiples of elementary charge, usually through electrons (because if you remove protons, the atoms themselves change).

Note: Elementary charges are elementary because they are the smallest stable charge. You can get quarks (with fractions of elementary charges), but will reform to these particles.

3 Sending and Receiving Charge

There are three methods for sending and receiving charge:

- Friction
- Conduction
- Induction

3.1 Friction

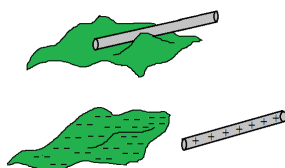


Figure 1: Example of Friction

When two objects rub with each other, the object with higher affinity will **take** electrons from the other. Usually, if you have two of the same object rub the same way, the objects have the same charge.

3.2 Conduction

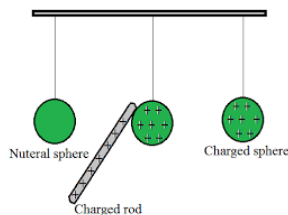


Figure 2: Example of Conduction

When two conductors (all objects are conductors in the end, if you try hard enough). If objects are identical, their charge will even out. Otherwise, they will just equalize. If I remember correctly, applying the "same" friction will cause the same charge result.

Note: These conductors equalize to minimize the system's *kinetic energy*.

3.3 Induction

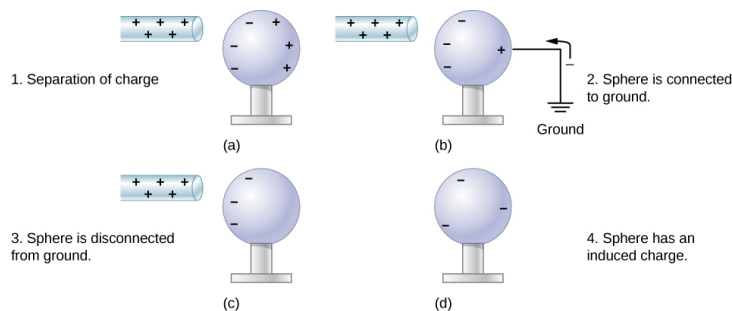


Figure 3: Example of Induction

A charge can change through proximity to an object. To equalize the charge afterwards, you can just tap the two objects together (if I remember correctly).

3.4 Note: Polarized Neutral

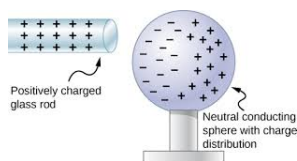


Figure 4: Example of a polarized neutral

A neutral object can be polarized through having the charged object attract the opposite charge of a neutral object, causing attraction through the distance.

3.5 Note: Question Stuff

The question worksheet was *Electric Charge and Methods of Charging*, and the only weird thing was question 4. To solve, just do

$$\frac{1.5 \text{ m} \times (6.023 \times 10^{23} \text{ particles per m})}{1 \times 10^6 \text{ particles per e}} \times \frac{1.6 \times 10^{-19} \text{ C}}{e} = 0.144 \text{ C}$$

6.023×10^{23} is Avogadro's (Avocado's) number, which is equal to the number of particles per mole. One just has to divide by that the number of particles before one is charged, and then to convert elementary charges to coulombs.

4 Coulomb's Law

Coulomb's Law is related to how two charged particles are attracted to each other.

4.1 Coulombs's Law from Regents

$$F_e = \frac{kq_1q_2}{R^2}$$

Problems with this formula:

- k assumes a vacuum
- q_1 , q_2 are point particles
- One has to assume one object is stationary

4.2 Building from Coulomb's Law for Physics C

1. We don't want to use k anymore. Instead we use $\frac{1}{4\pi\epsilon}$. ϵ represents the permittivity of a material ¹. For vacuums, the permittivity, or ϵ_0 , is equal to $8.85 \times 10^{-12} \frac{C^2}{Nm^2}$
2. To fix the fact that that q_1 and q_2 must be point particles, one will have to integrate over δq , or change in charge, to find the total charge, and integrate over δF . to find the force.

Putting everything together, we have the new formula:

$$\vec{F}_e = \frac{1}{4\pi\epsilon} \frac{q_1q_2}{R^2} \hat{r}$$

where $\hat{r}$ is vector pointing to the other charge, making it so that when the equation positive, the resultant vector is pointed away from the other particle. Remember, $\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$ [this is also on the reference table]

5 Electrical Fields

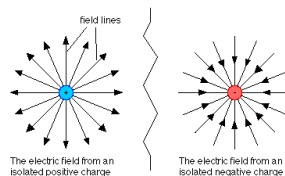


Figure 5: Examples of a Electrical Field

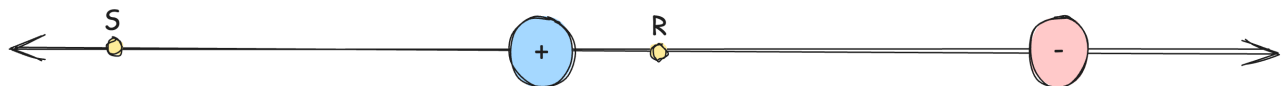
The formula for calculating a electric field, which is the force on the probe charge divided by the charge:

$$\vec{E}(r) = \frac{\vec{F}_0(r, q_0)}{q_0} = \frac{\frac{1}{4\pi\epsilon_0} \frac{Qq_0}{r^2}}{q_0} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} \frac{N}{C}}$$

NOTE: Always assume that the satellite particle is positive.

Example

The figure below shows a proton p and an electron e on an x-axis. What is the direction of field due to the electron at points S and R, and then the net charge.



¹the ability of a particle to hold energy at a specific field without breaking down

- The electron's field on S is towards the electron in a rightward direction, and at R is towards the electron in a left direction
- The net field at S is a lesser (lesser magnitude) arrow towards the right, and at R its a larger arrow (greater magnitude) to the left.

5.1 Calculating the Electrical Charge of Non-Point Particles

If we have a long thin rod, we have a few methods to try to integrate and find the field:

- You can integrate (**DRAWBACKS:** Both wierd directionally and changes with the charge too, not always ideal)
- Chickening out and using a ∞ lengthed rod, or use **Gauss' Law** (ohhhhh...)

6 Calculating Continuous Charge Distributions for a Ring

Reminder: It's arbitrarily difficult to calculate the charge of a continious shape. Therefore in order to solve these problems, one has to simplify stuff.

Simplification 1: The field exerted in point R by a charged ring.

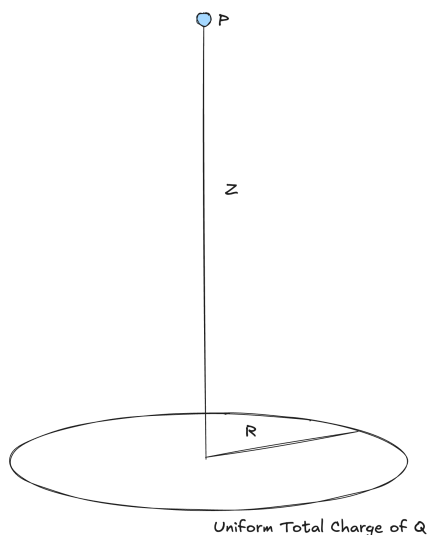


Figure 6: the Ring in question

Simplifications (it's a continious distance, but):

- Symmetry
- Rigid Structure

6.1 Deriving the Formula for Calculating the Electrical field

Let's divide this method into several steps

1. Define Charge Density
2. Use symmetry
3. Integration

Step 1: Defining Charge Density

Let's create a new value, and let's define it as charge density, λ , to help make this easier.

$$\lambda = \frac{\delta q}{\delta l}$$

Also, from the special case of it being a ring, we know that charge density is also the total charge minus the radius of the ring:

$$\text{Charge Density} = \frac{Q}{2\pi r}$$

Then, we know that:

$$\lambda = \frac{\delta q}{\delta l} = \frac{Q}{2\pi r}$$

$$\delta q = \lambda \delta l$$

Step 2: Finding $\delta \vec{E}$ from δq

Let's first start with integrating and then substituting for the change in charge.

$$\delta \vec{E} = \frac{1}{4\pi\epsilon R^2} \delta Q = \frac{1}{4\pi\epsilon} \frac{\lambda \delta l}{z^2 + r^2}$$

The visualization of the new figure:

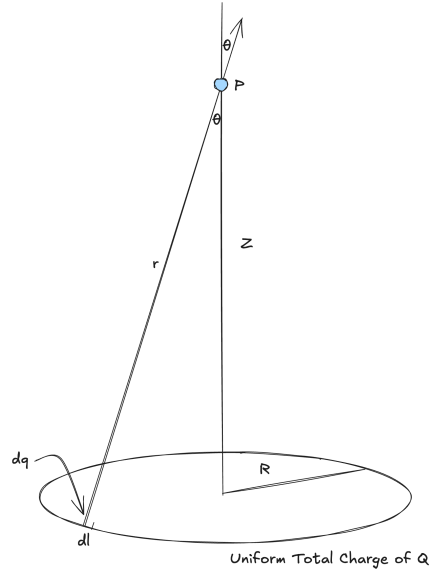


Figure 7: the Ring in question

Step 3: Use Symmetry

Common sense time: if you have two little $\delta \vec{F}$ from two little δq from two δl which are on opposite sides of circle, they cancel out. This is where we get $\delta \vec{E}_{\text{VERT}}$. After we integrate that, we should do some substituting

$$\begin{aligned} \delta E_{\text{VERT}} &= |\delta \vec{E}| \cos \theta = \frac{1}{4\pi\epsilon} \frac{\lambda \delta l}{(z^2 + R^2)} \cos \theta \\ &= \frac{1}{4\pi\epsilon} \frac{\lambda \delta l}{(z^2 + R^2)} \times \frac{z}{r} \left[\cos(\theta) = \frac{z}{r} \right] \\ &= \frac{1}{4\pi\epsilon} \frac{\lambda \delta l}{(z^2 + R^2)^{3/2}} z \end{aligned}$$

What's so great about this is that everything is constant, so integrating over the length of the ring would be trivial.

Step 4: Integrate

$$\begin{aligned} \vec{E} &= \int \delta E_{\text{VERT}} = \frac{1}{4\pi\epsilon} \frac{\lambda z}{(z^2 + R^2)^{3/2}} \int_0^{2\pi R} \delta l \\ &= \frac{\frac{Q}{2\pi R} z}{4\pi\epsilon(z^2 + R^2)} 2\pi R \left[\text{NOTE: } \lambda = \frac{Q}{2\pi R} \right] \\ &= \boxed{\frac{Qz}{4\pi\epsilon(z^2 + R^2)^{3/2}}} \end{aligned}$$

Some Test Cases

- $E_{z=0} = 0$
- $E_{z \gg r} = \frac{Q}{4\pi\epsilon z^2}$ [basically the point formula, its very far away it behaves like a point particle]

7 Calculating Continuous Charge Distributions for a Charged Circular Rod

To solve this problem, let's do a few things:

- Define Linear Charge Density, λ
- Pick infinitesimal charge element and define the infinitesimal field from the infinitesimal charge
- Use Symmetry to determine the component that wouldn't cancel
- Change variables as needed and integrate to find the field

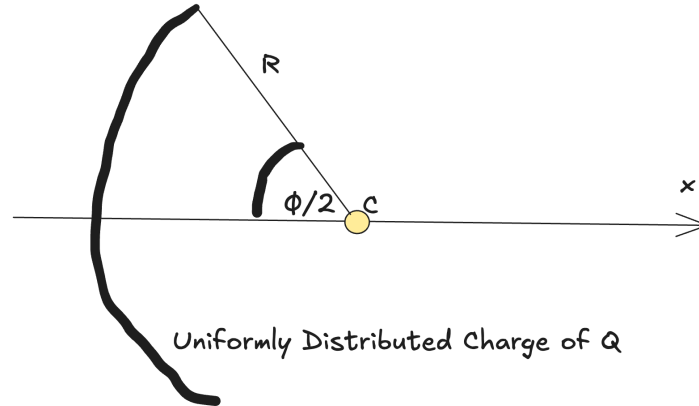


Figure 8: Example of a charged circular rod

Step 1: Defining Linear Charge Density

Let's try to find λ and its relations to angle

$$\begin{aligned}\lambda &= \frac{\delta Q}{\delta L} = \frac{Q}{2\pi r \frac{\phi}{2\pi}} = \frac{Q}{\phi R} \\ \delta Q &= \lambda \delta l \\ \delta L &= R \delta \theta \\ \delta Q &= R \lambda \delta \theta\end{aligned}$$

Step 2: Defining the Infinitesimal Field from the Infinitesimal Charge

$$\delta \vec{E} = \frac{1}{4\pi\epsilon R^2} \delta Q = \frac{1}{4\pi\epsilon} \frac{Q}{\phi R} \delta l$$

Step 3: Use Symmetry to Uncancelable Component

$$\delta \vec{E}_{\text{NC}} = \delta \vec{E} \cos(\theta) = \frac{1}{4\pi\epsilon} \frac{Q}{\phi R} \cos(\theta) \delta l$$

Step 4: Change Variables and Integrate

$$\begin{aligned}\delta \vec{E}_{\text{NC}} &= \frac{1}{4\pi\epsilon R^2} \frac{Q}{\phi R} \cos(\theta) \delta l = \frac{1}{4\pi\epsilon R^2} \cos(\theta) \frac{Q}{\phi R} R \delta \theta \\ \vec{E} &= 2 \int_0^{\frac{\phi}{2}} \frac{1}{4\pi\epsilon R^2} \frac{Q}{\phi R} R \cos(\theta) \delta \theta \\ &= \boxed{\frac{Q}{4\pi\epsilon R^2 \phi} 2 \sin\left(\frac{\phi}{2}\right)}\end{aligned}$$

8 Find the Electrical Field of a Disk

Find the electrical field of a point a distance z above a uniformly charged disk of radius R , with area A .

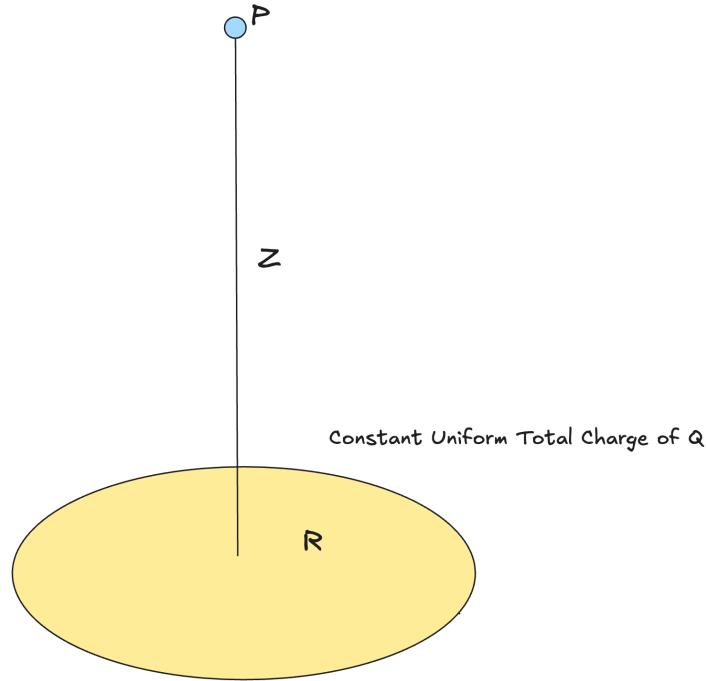


Figure 9: Example of a charged disk

Step 1: Find charge density

σ is the charge density

$$\sigma = \frac{\delta q}{\delta A} = \frac{Q}{\pi R^2}$$

$$\delta q = \sigma \delta A$$

Let's now split the disk into rings of radius r ($0 < r < R$), thickness δr and charge $\delta q = \sigma \delta A$

$$\delta A = 2\pi r \delta r$$

$$\delta q = \sigma \delta A = 2\pi r \sigma \delta r$$

Step 2: Finding an infinitesimal field from infinitesimal charge

$$\delta \vec{E} = \frac{z \delta q}{4\pi\epsilon(z^2 + r^2)^{\frac{3}{2}}} = \frac{z\sigma}{4\epsilon} \frac{2r\delta r}{(z^2 + r^2)^{\frac{3}{2}}}$$

Step 3: Integrate

$$\text{Let } u = z^2 + r^2; \delta u = 2r\delta r$$

$$\begin{aligned} \vec{E} &= \frac{z\sigma}{4\epsilon} \int_0^R \frac{zr}{(z^2 + r^2)^{\frac{3}{2}}} \delta r \\ &= \frac{z\sigma}{4\epsilon} \int_{z^2}^{z^2+R^2} u^{-\frac{3}{2}} \delta u = -\frac{z\sigma}{4\epsilon} \left[u^{-\frac{1}{2}} \right]_{z^2}^{z^2+R^2} \\ &= -\frac{z\sigma}{4\epsilon} \left[\frac{1}{\sqrt{z^2 + R^2}} - \frac{1}{z} \right] = \frac{z\sigma}{4\epsilon} \left[\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}} \right] \\ &= \frac{\sigma}{2\epsilon} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right] = \boxed{\frac{Q}{2\pi R^2\epsilon} \left[1 - \frac{z}{\sqrt{z^2 + R^2}} \right]} \end{aligned}$$

Some Checks Just some limits tomfoolery

$$E \left(\begin{smallmatrix} z \rightarrow \infty \\ R \rightarrow \infty \end{smallmatrix} \right) = \frac{\sigma}{2\epsilon}$$

9 Constant Electric Field

Constant electric fields are just that. They're constant, and they apply to particles in interesting ways, very similar to gravity.

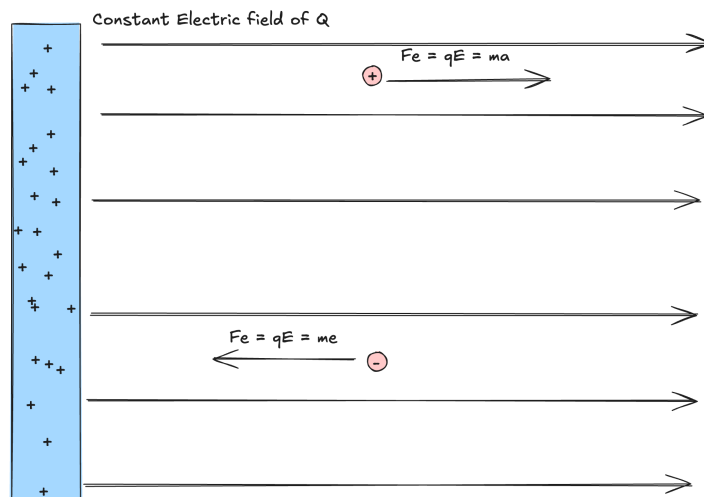


Figure 10: Example of a constant electric field

You get parabolic stuff too, if apply an initial velocity.

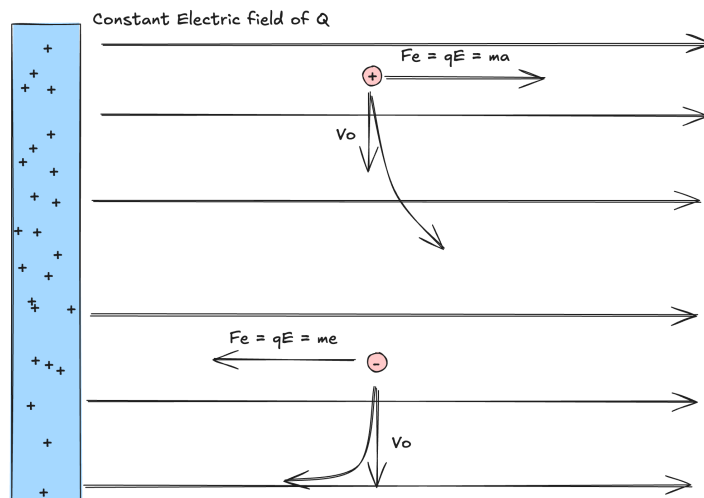


Figure 11: Example of a constant electric field with an initial velocity

10 Electrical Potentials

Electric potentials are measures of the potential energy created by an electric field. You can think of them as such:

Potential: Potential Energy :: Field: Force Energy

For the potential energy of point charges, there is a very big positional element here. As long as you have two charged particles, as long as they want to move somewhere, there will always be potential energy in that system. Usually, systems want to reduce that energy to zero.

10.1 Formulas and Dimensional Analysis

Let's look at the formulas for volts and field, and see what we can make for them. Let's have the formula for volts, and put of the formula for electric field for comparison.

$$V = \frac{U}{q}$$

$$\vec{E} = \frac{\vec{F}}{q}$$

Let's do a quick dimensional analysis too for these *volts*:

$$[V] = \frac{[J]}{[C]}$$

Differences between field and potential:

- Field is a vector quantity, voltage is a scalar quantity
- Field lines are always perpendicular across vector lines

Let's have a visualization changes of potential over time:

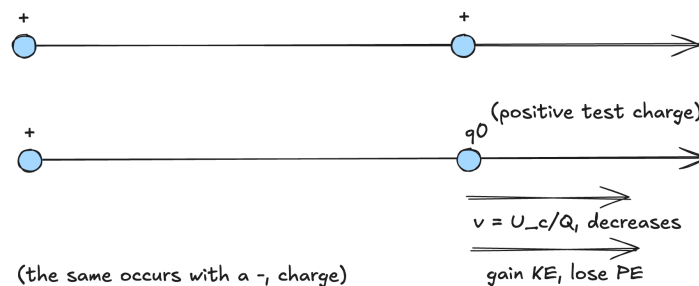


Figure 12: The change of charge across a number line

Notes:

- Voltage decreases as one goes through a field lines

10.2 Equipotentials

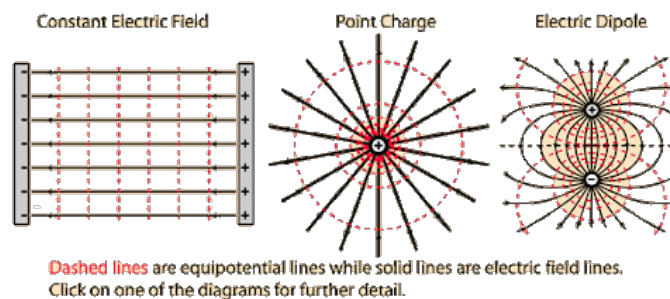


Figure 13: Example of electric fields vs electric potentials

- Lines across these show the areas of equidistant potentials
- Circular around point charge
- *Perpendicular to field lines*, (e.g. Electric field is the negative gradient of potential), or $E = -\Delta V$

10.3 Calculating Electrical Potential from Electric Field

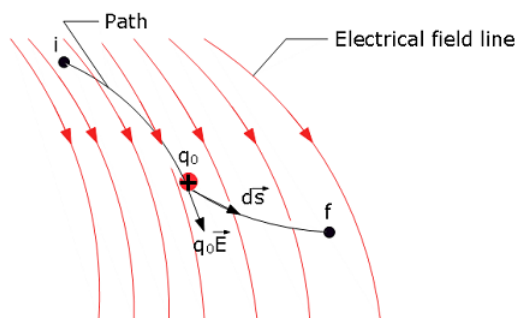


Figure 4-3 A test charge q_0 moves from point i to point f along the path shown in a nonuniform electric field. During a displacement $d\vec{s}$, an electric force $q_0\vec{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

Figure 14: Visualization of the different differentials here

To try to find a formula for this, starting from the givens and working up.

$$\vec{F} = q_0\vec{E}$$

$$W = \int \delta W$$

$$W = \int \delta w = \int_i^f \vec{F} \delta \vec{s}$$

$$= -\Delta U = -q_0 \Delta V$$

$$\Delta v = \boxed{-\int_i^f \vec{E} \delta s}$$

Given that electric field is a vector, and potentials are scalars, one might have to treat these as partial derivatives of x and y . So we have the original equation:

$$\vec{E} = \frac{Q}{4\pi\epsilon R^2} = -\frac{\delta V}{\delta s}$$

But we have to split it into components:

$$E_x = \frac{\delta V}{\delta x}$$

$$E_y = \frac{\delta V}{\delta y}$$

$$\vec{E} = -\frac{\Delta V}{\Delta S} = \frac{\delta V}{\delta x} + \frac{\delta V}{\delta y}$$